Roll Number: _____

Thapar Institute of Engineering and Technology (TIET), Patiala
Department of Computer Science & Engineering (CSED)
AUXILIARY EXAMINATION

B.E.	**Course Code**: UCS802 **Course Name**: Compiler Construction
August 22, 2025	Friday, 5:30 PM
Time: 3Hrs, Max. Marks: 100	**Name of Faculty**: Dr. Shashank Singh

Note: Attempt any 5 questions. Assume missing data, if any suitably.

Q.N.	Question	Marks	CO	BL

Q1 (a)	**Apply steps of Compilation on following HLL code snippet and find the final output Machine code. Explain every phase in detail.**
	long simple(long a, long b, long c) {
	return (a - b) + (c << 1);
	}
		10	CO1	BL3

(b)	**Find total number of tokens in following Code**
	#include <stdio.h>
	int main() {
	// Variable declarations
	int x = 10, y = 20, sum = 0;
	/* Compute sum using loop */
	for(int i = 0; i < 5; i++) {
	if(i % 2 == 0) {
	sum = sum + x * y;
	} else {
	sum = sum - x;
	}
	}
	Printf ("Final Sum = %d\n", sum);
	return 0;
	}
		10	CO1	BL2

Q2 (a)	**Explain left recursion and left factoring with example. Why it is essential to remove it before applying top-down parsing.**	10	CO2	BL2
(b)	**Construct the LL(1) parsing table for following grammar** E -> E+T/T T-> T*F/F F-> id/ (E) **Also, parse the string id+id*id using LL(1) Parser.**	10	CO2	BL6
Q3 (a)	**Construct the LR(0) parsing table for following grammar** S-> AA A-> aA/b	10	CO2	BL6

	and check how many SR and RR conflicts are present in the table.			
(b)	**Find first() and follow() for all the non-terminals in the following grammar.** S->Aa/aAc/Bc/bBa A-> d B-> d	**10**	**CO2**	**BL2**
Q4 (a)	**Generate the three-address code for following C Program using batch patching technique** Main() { int i = 1; Int a[10]; While (i<=10) a[i] =; }	**10**	**CO3**	**BL3**
(b)	**A shift reduce parser is applied on the following SDT** S -> AS {printf (1)}; | AB {printf (2)}; A -> a {printf (3)}; B -> bC {printf (4)}; | dB {printf (5)}; C -> c {printf (6)}; **For the input string 'aaadbc' find out the output.**	**10**	**CO3**	**BL2**
5 (a)	**(i) Differentiate between lexeme, token and pattern** **(ii) What are the issues in lexical analysis phase**	**10**	**CO1**	**BL4**
(b)	**Construct NFA to recognize the language (a/b)*ab using Thomson construction method and minimize it using subset construction.**	**10**	**CO1**	**BL6**
6 (a)	**Explain various code optimization techniques with the help of suitable examples.**	**10**	**CO4**	**BL1**
(b)	**Compare and contrast Static, Stack, and Heap Allocation with suitable examples.**	**10**	**CO3**	**BL4**